

Ghadi Nehme

 ghadi-nehme ·  Google Scholar ·  Portfolio ·  ghadi@mit.edu

Education

Doctor of Philosophy in Artificial Intelligence | *MIT* Sep 2024 – Present
GPA: 5.0/5.0. Awarded the Jamieson Presidential Fellowship (2024). **Advisor:** Prof. Faez Ahmed.

Master of Science in Mechanical Engineering | *Stanford University* Sep 2022 – Jun 2024
GPA: 4.0/4.0. **Specialization:** Robotics, Design and Artificial Intelligence.
Courses: Machine Learning, Computer Vision, Decision Making Under Uncertainty, Principles of Robot Autonomy.

Bachelor of Science in Mathematics & Physics | *Ecole Polytechnique, France* Sep 2019 – Jun 2022
GPA: 4.22/4.0, Class Valedictorian, Summa Cum Laude mention.
Minor: Biology. **Organizations:** President of the Robotics club, Mathematics club, and Arts club.

Exchange Semester | *University of Toronto* Sep 2021 – Dec 2021
GPA: 4.0/4.0. **Engineering courses:** Mechanical Engineering Design, Circuit Analysis, Solid Mechanics.

Summer School: Introduction to Artificial Intelligence | *CentraleSupélec* Jul 2021

Awards

- Recipient of the GE Vernova graduate fellowship, 2026
- Best Paper Award for the AI Track and Overall Best Paper Award for the ASME's IDETC-CIE 2026 conference
- Recipient of the MIT Thomas Sheridan Award, which is awarded to an undergraduate or graduate student for creativity in human-machine integration, 2026
- Outstanding Student Distinction, Ecole Polytechnique: “distinguished himself through his dedication to the École Polytechnique community and/or to the welfare of society,” 2019–2022.
- Bronze medalist at the International Mathematics Competition, 2021.
- Recipient of the “Excellence Scholarship” from the École Polytechnique Foundation, 2020.
- Recipient of the highly selective “Excellence-Major Scholarship” from the French Government, 2019.
- First prize in the Science Fair organized by the USJ, 2018, for building an autonomous delivery car.

Publications & Research Projects

CADFit: Precise Mesh-to-CAD Program Generation with Hybrid Optimization (*ICML 2026*) | *MIT* Oct 2025 – Jan 2026

- Developed CADFit, an AI-guided optimization method for mesh-to-CAD reconstruction, enabling editable outputs with operations such as extrusions, revolutions, booleans, fillets, and chamfers.
- Achieved state-of-the-art reconstruction accuracy on DeepCAD, Fusion360, and ABC benchmarks, including an end-to-end image-to-CAD pipeline.

LAMP: Data-Efficient Linear Affine Weight-Space Models for Parameter-Controlled 3D Shape Generation (*ICML 2026*) | *MIT* May 2025 – Sep 2025

- Proposed LAMP, a data-efficient framework for interpretable, parameter-controlled 3D mesh generation via affine mixing in aligned SDF weight space.
- Enabled accurate interpolation and safe extrapolation up to 100% beyond training ranges using only 100 exemplars.
- Introduced a linearity-mismatch safety metric and demonstrated performance-driven aerodynamic optimization on DrivAerNet++ and BlendedNet. [\[ArXiv\]](#)

VideoCAD: A Dataset and Model for Learning Long-Horizon 3D CAD UI Interactions from Video (*NeurIPS 2025*) | *MIT* Jan 2025 – May 2025

- Developed VideoCAD, the first large-scale dataset of over 41K annotated videos capturing CAD user interface interactions, enabling long-horizon learning of complex 3D design workflows.
- Trained VideoCADFormer, a transformer model achieving state-of-the-art performance in CAD UI action prediction.
- Introduced VideoCAD VQA, a benchmark with 1,200 visual questions to assess multimodal LLMs' 3D spatial reasoning and video understanding, exposing limitations of models in precision engineering tasks. [\[ArXiv\]](#)

Sketch-to-CAD: Generating CAD Models from Hand-Drawn Sketches | *MIT* Sep 2024 – Dec 2024

- Built a dataset with 166,255 hand-drawn sketches and 3D models of parts from the DeepCAD dataset.
- Developed a Sketch-to-CAD framework using a CNN encoder and Transformer decoder leveraging transfer learning.
- Achieved high accuracy on the DeepCAD dataset, demonstrating effective geometric clustering in latent space for generating CAD models from hand-drawn sketches. [\[Report\]](#)

LLM-Driven Adaptive Robotic Manipulation for Complex Tasks | *MIT* Sep 2024 – Dec 2024

- Developed an LLM-driven robotic system for high-level task planning in a simulated kitchen using VLMs.
- Designed a modular execution framework enabling a robotic arm to perform stacking, sorting, and table setup tasks.
- Optimized grasping in object manipulation tasks using bounding boxes and antipodal methods. [\[Report\]](#)

Multi-Objective Optimization Framework for Robot Design | *Stanford University* Apr 2024 – Jun 2024

- Developed a multi-objective optimization framework using genetic algorithms and gradient descent to optimize robotic design for efficiency and safety in medical applications.
- Validated the framework with case studies, achieving enhanced manipulability and precision in robots designed for complex surgical and dental tasks. [\[Report\]](#)

Enhancing Robotic Manipulation: Multi-Task RL and Single Life RL in Meta-World | *Stanford University* Apr 2023 – Jun 2023

- Developed MT-QWALE, a multi-task RL algorithm enabling robotic arms to complete MetaWorld tasks in a single trial under distribution shifts.
- Outperformed MT-SAC, achieving near-100% success under randomized object positions.
- Conducted an ablation study by masking the end goal position, finding that MT-QWALE can still accomplish tasks relying solely on reward feedback. [\[ArXiv\]](#) · [\[Code\]](#)

Safe Navigation: Training Autonomous Vehicles using Deep RL | *Stanford University* Jan 2023 – Mar 2023

- Developed deep RL models using DQN to enable autonomous vehicles to navigate urban environments safely in the CARLA simulator; achieved 94% success rate on unseen test trajectories.
- Implemented sensor data preprocessing of segmentation and depth images to extract relevant state information and reduce state space complexity. [\[ArXiv\]](#) · [\[Code\]](#)

Research and Work Experience

Graduate Research Assistant | *DeCoDe Lab, MIT* Jan 2025 – Present

- Conducting research on AI for multimodal CAD generation, developing large-scale datasets and models for parametric design generation, UI action learning, and precision-controlled optimization.

PhD Research Scientist Intern in AI for Product Design | *Meta* May 2026 – Aug 2026

- Built a multi-agent AI system that automates mechanical engineering workflows including CAD drawing inspection, parametric model generation, 2D drawing automation, and FEA surrogate modeling.
- Developed an AI-native VR design application where engineers interact through voice commands and hand gestures to generate 3D geometry, run finite element simulations, and navigate assemblies in an immersive environment.
- Trained ML surrogate models for rapid forward and inverse display design, enabling real-time performance prediction and design optimization.

Graduate Teaching Assistant | *MIT* Sep 2025 – Dec 2025

- 2.155: AI and ML for Engineering Design: Organized office hours and mentored students in their final projects.

Graduate Research Assistant | *CHARM Lab, Stanford University* Sep 2023 – Apr 2024

- Designing the system and control of a neurosurgery robot combining an endoscope, exoscope, and robotic arms.
- Training the Da Vinci robot to achieve the retraction of tissues of various stiffnesses autonomously.

Graduate Teaching Assistant | *Stanford University* Sep 2023 – Jun 2024

- MATH 104: Applied Matrix Theory: Organized weekly office hours and review sessions for more than 100 students.

Hardware Engineering Intern | *Google* Jun 2023 – Sep 2023

- Performed reliability competitor benchmarking study of tablets. Conducted failure analysis and formulated actionable design guidelines such as water ingress resistance and battery to carrier clearance design guideline.
- Analyzed the correlation between reliability robustness and system integration features.

Co-Founder & CTO | *DaVinci, AI startup for automating CAD* Oct 2023 – Jun 2024

- Built an AI model automating 2D sketch to CAD conversion, optimizing design processes.

Design Research Engineer | *Stanford University* Sep 2022 – Jun 2023

Supervisors: Dr. Mark Cutkosky, Dr. George Toye [\[Report\]](#)

- Five-member team project in collaboration with Johnson & Johnson for one academic year.
- Designed and built a smart pharmacy which dispenses single-packaged pills based on orders sent from a UI, using a Raspberry Pi. The goal is to reduce expired medications.

Graduate Research Assistant | *School of Medicine, Stanford University*

Jan 2023 – Jun 2023

Supervisor: Dr. Keren Haroush

[\[Report\]](#)

- Built a real-time choice decoding of the decision of a monkey playing at iterative Battle of Sexes using neural spikes.
- Achieved 90% accuracy on decoding monkey decisions by training a regression model on the neural data, using Python.

Robotics Research Intern | *Laboratory of Optics and Biosciences, CNRS*

Apr 2022 – Jun 2022

- Developed a robot for an organ-on-a-chip platform to renew the cell culture media every 3 hours.
- Built a CAD model of the robot, 3D printed the parts, tested the efficiency with fluid mechanics calculations.
- Integrated electronic control system based on Arduino, touch screen and servo-motors. [\[Report\]](#) · [\[Code\]](#)

Bachelor Thesis: Research Engineer Intern | *Solid Mechanics Lab, CNRS*

Jan 2022 – Mar 2022

- Characterized experimentally a 3D-printed polymer, designed the setup using SolidWorks.
- Tested the polymer using a hydraulic press, DIC and post-processing analysis using Python.
- Fitted the experimental data with a model for hyper-elastic elastomers.

Research Data Scientist Intern | *Dynamic Meteorology Lab, CNRS*

Jun 2021 – Aug 2021

- Identified the characteristics of gravity waves from balloons that traveled above Antarctica for months using Python.
- Developed a metric to select the closest trajectory of a virtual balloon to a real one for each of the 13 real balloons.

Skills

Software Python, C++, OOP, MATLAB, Julia, Git, TensorFlow, PyTorch, SolidWorks (CSWA), AutoDesk Fusion 360, Onshape, MuJoCo, ROS

Hardware Arduino, Raspberry Pi, 3D Printing, Prototyping, Hand Drafting

Leadership & Volunteering

General Secretary | *L'ORE: Student's Government of the Bachelor Program*

Jun 2020 – Jun 2021

- Led a team of nine members, organized monthly social events, collected funds for the construction of a water fountain in Laos in collaboration with the organization Safe Water Cube.

Team Leader | *SSCC Missionary Group*

Jun 2017 – Jun 2022

- Organized activities for children and elderly in need, helped people affected by the August 4th explosion in Beirut.

Languages & Hobbies

Languages French, English, Arabic, Japanese (Limited)

Hobbies Drawing, Painting, Playing the piano, Soccer, Tennis, Squash, Pickleball, Prototyping